Relational Algebra

What is Relation?

- I am sure that you are familiar with many relations such as “less than,” “is parallel to,” “is a subset of,” and so on. In a certain sense, these relations consider the existence or nonexistence of a certain connection between pairs of objects taken in a definite order.
- Formally, we define a relation in terms of these “ordered pairs.”
- An *ordered pair* of elements a and b, where a is designated as the first element and b as the second element, is denoted by (a, b). In particular,
  - $(a, b) = (c, d)$
- if and only if $a = c$ and $b = d$. Thus $(a, b) = (b, a)$ unless $a = b$. This contrasts with sets where the order of elements is irrelevant; for example, {3, 5} = {5, 3}.

Product Sets

- Consider two arbitrary sets A and B. The set of all ordered pairs (a, b) where $a \in A$ and $b \in B$ is called the *product*, or *Cartesian product*, of A and B. A short designation of this product is $A \times B$, which is read "A cross B." By definition,
  - $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$
- Example: Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then
- $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$
- $B \times A = \{(a, 1), (b, 1), (c, 1), (a, 2), (b, 2), (c, 2)\}$
- Also, $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

(Here $A\times B \neq B \times A$)

Number of Elements in Product Sets

In our previous example, $n(A \times B) = 6 = 2(3) = n(A)n(B)$

- We can say $n(A\times B) = n(A)n(B)$

Relations

- Let A and B be sets. A *binary relation* or, simply, *relation* from A to B is a subset of $A \times B$.
- Suppose R is a relation from A to B. Then R is a set of ordered pairs where each first element comes from A and each second element comes from B. That is, for each pair $a \in A$ and $b \in B$, exactly one of the following is true:
  1. $(a, b) \in R$; we then say "a is R-related to b", written aRb.
  2. $(a, b) \notin R$; we then say "a is not R-related to b", written aRb.
- If R is a relation from a set A to itself, that is, if R is a subset of $A^2 = A \times A$, then we say that R is a relation *on* A.
- The *domain* of a relation R is the set of all first elements of the ordered pairs which belong to R, and the *range* is the set of second elements.

Example:

- $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$, and let $R = \{(1, y), (1, z), (3, y)\}$. Then R is a relation from A to B since R is a subset of $A \times B$. With respect to this relation, $1Ry$, $1Rz$, $3Ry$, but $1 \not R x, 2 \not R x, 2 \not R y, 2 \not R z, 3 \not R x, 3 \not R z$
  - The domain of R is $\{1, 3\}$ and the range is $\{y, z\}$
- Set inclusion $\subseteq$ is a relation on any collection of sets. For, given any pair of set A and B, either $A \subseteq B$ or $A \nsubseteq B$.
- A familiar relation on the set **Z** of integers is "m divides n." A common notation for this relation is to write $m|n$ when m divides n. Thus $6 \mid 30$ but $7 \nmid 25$

- Consider the set *L* of lines in the plane. Perpendicularity, written "⊥," is a relation on *L*. That is, given any pair of lines *a* and *b*, either $a \perp b$ or $a \not\perp b$. Similarly, "is parallel to," written "||," is a relation on *L* since e$a \parallel b$ or $a \nparallel b$.
- Let *A* be any set. An important relation on *A* is that of *equality*, {*(a*, *a)* | $a \in A$} which is usually denoted by "=." This relation is also called the *identity* or *diagonal* relation on *A* and it will also be denoted by *A* or simply .
- Let *A* be any set. Then $A \times A$ and ∅ are subsets of $A \times A$ and hence are relations on *A* called the *universal relation* and *empty relation*, respectively.

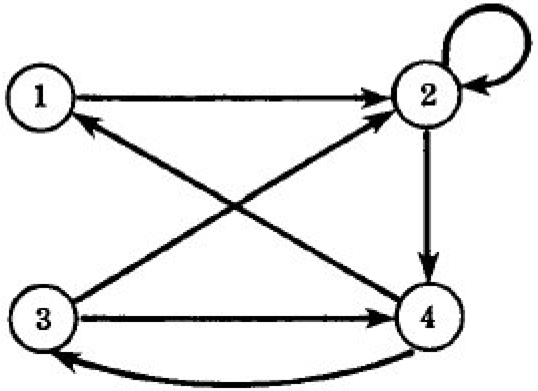
Inverse Relation

- Let *R* be any relation from a set *A* to a set *B*. The *inverse* of *R*, denoted by R^{-1}, is the relation from *B* to *A* which consists of those ordered pairs which, when reversed, belong to *R*; that is,
  - $R^{-1} = \{(b, a) \mid (a, b) \in R\}$
- For example, let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Then the inverse of
  - $R = \{(1, y), (1, z), (3, y)\}$ is $R^{-1} = \{(y, 1), (z, 1), (y, 3)\}$
- Clearly, if *R* is any relation, then $(R^{-1})^{-1} = R$. Also, the domain and range of R^{-1} are equal, respectively, to the range and domain of *R*. Moreover, if *R* is a relation on *A*, then R^{-1} is also a relation on *A*.

Pictorial Representation of Relation

Directed Graph:

- Say relation R on the set $A = \{1, 2, 3, 4\}$: $R = \{(1, 2), (2, 2), (2, 4), (3, 2), (3, 4), (4, 1), (4, 3)\}$



Pictures of Relations on Finite Sets

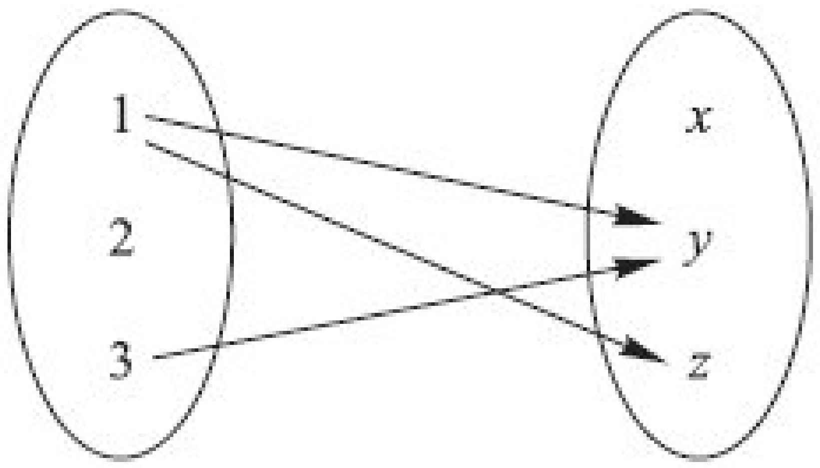
Suppose *A* and *B* are finite sets. There are two ways of picturing a relation *R* from *A* to *B*.

1. Form a rectangular array (matrix) whose rows are labeled by the elements of *A* and whose columns are
   I. labeled by the elements of *B*. Put a 1 or 0 in each position of the array according as $a \in A$ is or is not
   II. related to $b \in B$. This array is called the *matrix of the relation*.

	x	y	z
1	0	1	1
2	0	0	0
3	0	1	0

(i)

1. Write down the elements of *A* and the elements of *B* in two disjoint disks, and then draw an arrow from
    I. $a \in A$ to $b \in B$ whenever *a* is related to *b*. This picture will be called the *arrow diagram* of the relation.



    II. A = {1, 2, 3}, B = {x, y, z} and R = {(1, y), (1, z), (3, y)}